

Cristian C. Spagnuolo

NLP ENGINEER

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Summary

NLP engineer focused on production LLMs - efficient inference, fine-tuning (SFT, DPO, GRPO), and reliable evaluation. Strong software-engineering foundation: clean, maintainable code, thorough testing, and CI/CD.

Experience

Artificially SA

Lugano, Switzerland

NLP ENGINEER

05/2025 - Present

- Optimized on-premise LLM inference with quantization and **TensorRT/vLLM** runtimes: **4x less GPU memory, 1.8x faster**.
- Main contributor to the internal **fine-tuning framework (SFT, DPO, GRPO)**, multi-GPU training, and **MLflow** tracking.
- Built an **LLM evaluation pipeline** (retrieval, metrics, artifacts, non-regression tests), cutting **eval effort by 66%**.
- Engineered a **tabular-data Q&A agent** with secure code generation, visualization, and forecasting; **>90% accuracy**.
- Authored an **automatic API-doc generator** for the internal Python template, producing **MkDocs** pages and removing manual upkeep.

ML cube

Milano, Italy

AI/ML ENGINEER

09/2024 - 05/2025

- Trained a **pricing agent** with PPO for hotel booking, increasing simulated revenue by **16%** over baseline strategies.
- Created an **LLM-powered SQL agent** for financial users, enabling **natural language queries** with **91% accuracy**.
- Designed an **AI regulatory assistant** on knowledge graphs, improving **analyst decision accuracy by 38%**.
- Enhanced the "ML cube Platform" with segmented monitoring for drift-detection, raising **detection coverage by 35%**.
- Automated **document digitization** through a pipeline for structured data extraction, achieving **87% OCR accuracy**.

Achievements

COMPETITIONS

2025 **2nd Place**, Hackapizza 🏆

IBM Studios, Milano

CERTIFICATIONS

2024 **Certifications, Machine Learning** IBM 🏆 | **Advanced Data Analytics** Google 🏆

Milano, Italy

Education

Politecnico di Milano

Milan, Italy

MASTER'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING – **110/110 CUM LAUDE**

02/2020 – 07/2024

Università degli Studi del Sannio

Benevento, Italy

BACHELOR'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING – **110/110 CUM LAUDE**

09/2015 – 10/2019

Projects

LLMs from Scratch 🏆

PROJECT

07/2025 - 10/2025

- Implemented **GPT-2** from scratch in **PyTorch**: tokenization, embeddings, multi-head attention, and training loop.
- Leveraged **LLM scaling** insights to guide architecture and hyperparameters, cutting training time by **10%**.
- Technologies**: Python, PyTorch, NumPy, Pydantic, Uv, Ruff, MLflow

Image Distribution Drift Simulator 🏆

PROJECT

10/2024 - 12/2024

- Built an **image drift simulator** to reproduce **synthetic distribution shift**, improving AI model robustness testing by **22%**.
- Technologies**: Python, Polars, Torch, Plotly, TorchVision, PIL.

Curriculum Learning for Reinforcement Learning – Politecnico di Milano

Advisor: Prof. Marcello Restelli

MASTER'S THESIS

02/2023 - 07/2024

- Designed a novel **Curriculum Learning** framework for **Reinforcement Learning** built on **Policy Gradient Methods (PPO)**, deriving algorithms that progressively scale task difficulty to solve sparse-reward problems where standard RL fails.
- Technologies**: Python, PyTorch, NumPy, Gymnasium, PPO, Policy Gradient Methods.

Skills

Programming

Python, Java, C, C++, Assembly, SQL

AI/ML

Pandas, Torch, Polars, NumPy, TensorFlow, LangChain, LangGraph, Hugging Face, Ollama, DeepSeek, OpenAI, MLflow, scikit-learn, Matplotlib, Plotly, Gemini, Vertex AI, Gymnasium, Mistral, Qwen, vLLM, TensorRT

Technologies

GCP, AWS, Azure, Git, GitHub, Docker, Docker Compose, Traefik, MongoDB, FastAPI, Firebase, Django, Node-RED, Make, Uv, Neo4j

Languages

English, Italian